Machine Learning
Project-Phase 3

In this phase we will focus on feature engineering of our selected datasets.

1) **How to do feature engineering effectively?**
   - **a)** Use 5 different algorithms to find important features in your dataset. SHAP and LIME can also be used to explore importance of features.
   - **b)** Create new meaningful features from your existing dataset.
   - **c)** You can use any tree base method to see how well your new features are performing.
   - **d)** After completing feature engineering perform Standardization / Normalize as required.
   - **e)** At the end see if K-means can be used to create any new feature?

Note:

1) In this phase you don't need to fine tune parameters of algorithms. Use LightGBM with its default settings to check performance of new features. LightGBM is fast and it will save your time. Running LightGBM on kaggle's GPU will speed up things further and save more time.
2) Only keep the features which are meaningful and help improve the results. Also try to drop less important features and see if model performs better or not.
3) For creating new features don't rely only on important features. Less important features can also be used.
4) In your notebook explain the significance of new features which you have created. Also write why they were required. If you drop any feature write its reason.
5) Once you have finished save your data as a csv file

Best Wishes